Majid Baradaran Akbarzadeh

baradara@msu.edu, (517) 980-4296, East Lansing, MI, USA
Scholar LinkedIn

EDUCATION

Michigan State University — **East Lansing, MI**
PhD, Mechanical Engineering, *34 credit hours passed*, GPA: 3.94/4.0 — Sep. 2022-Present
Selected Coursework: Linear Systems and Control, Advanced Dynamics, Nonlinear Vibration, Nonlinear Dynamics and Chaos, Foundation of Data Science, Machine Learning
Research Topic: Energy harvesting For Different Applications: Fluid Induced Vibration and Vehicle Suspension
(Project funded by Office of Naval Research (ONR)) and (National Science Foundation (NSF))

Ferdowsi University of Mashhad — **Mashhad, Iran**
M.Sc., Mechanical Engineering, *32 credit hours passed*, GPA: 4.0/4.0 — Sep. 2020
Selected Coursework: Fuzzy Control, Theory of Perturbation, Energy Harvesting and Modal Analysis
Research Topic: Nonlinear dynamic modeling of parallelogram flexure

Azad University of Mashhad — **Mashhad, Iran**
B.Sc., Mechanical Engineering*, 142 credit hours passed*, GPA: 4.0/4.0, **1st** rank — Aug. 2016
Final project: Simulation of car crash test in Ansys software

RESEARCH WORK EXPERIENCE

Michigan State University - Dynamics and Vibration Lab — **East Lansing, MI**
Graduate Student Research Assistant – Advisors: Dr. Wei Che Tai and Dr. Ranjan Mukherjee — Jan. 2023-Present

Projects:

- Fluid conveying pipe system energy harvesting (internal flow system)
- Studied electrically interconnected car suspension with **resistive load** under random excitations for energy harvesting
- **RLC** two port electrical network **optimization** for **energy harvesting** and **ride comfort** from vehicle suspension
- Studied vehicle dynamics and electrical network optimization (**time** and **frequency** analysis)
- Participated in doing experiment on **full-scale shock absorber** of the vehicle (Acknowledged in the paper)

Ferdowsi University of Mashhad - Precision, Vibration, And Control Laboratory — **Mashhad, Iran**
Graduate Student Research Assistant — Jun. 2017-Sep. 2021

Projects:

- Studied **nonlinear** dynamics of **parallelogram beam flexure** under harmonic base excitation (MSc thesis)
- **Nonlinear controlling** of ball and plate system (BPS) based on **Fuzzy** logic
- Nonlinear analysis of cantilever beam with tip-mass under **axial** and **transversal** harmonic loads **energy harvesting** formulation and **modal experiment**

Teaching Assistantship:

- Mechanics of Materials I & II (B.Sc. program)
- Statics (B.Sc. program)
- Differential Equations (B.Sc. and M.Sc. programs)
- Mechanical Vibrations (B.Sc., M.Sc., and PhD programs)
- Mechanical Design (PhD program)
- Vibrations and Control Lab (PhD Program)

RESEARCH INTERESTS

- Nonlinear dynamics, vibration, and control analysis
- Structural dynamics
- Fluid structure interaction (FSI)

- Electromechanical systems analysis
- Vehicle dynamics
- Electrical network optimization and analysis
- Energy harvesting of electromechanical systems
- Multibody dynamics and vibration
- Classical and Fuzzy control logics

SKILLS

CAD and CAE: SOLIDWORKS, ANSYS, ABAQUS
Programming: MATLAB/Simulink, Wolfram Mathemtaica, Maple, Python
Hardware development & testing: Basic machining and manufacturing techniques, Experimental setup and testing, Vibration analysis and modal testing, MTS machine testing
Research skills: Analyzing Electromechanical Systems, Vehicle Dynamics, Electrical Network Analysis, Energy Harvesting from Electromechanical Systems, Fluid Structure Interaction

SELECTED PUBLICATIONS

1) **M. Baradaran Akbarzadeh**, H. Moeenfard, S. Awtar "*Nonlinear Dynamic Modeling of a Parallelogram Flexure*" (Paper accepted and published in *Mechanism and Machine Theory journal*) Link
2) **M. Baradaran Akbarzadeh**, W.C. Tai "*Analysis of an Electrically Interconnected Half-Car Suspension System With Resistive Loads Under Random Road Excitation*" (Paper accepted and published in *AMSE's IDETC-CIE 2024 conference*) Link
3) **M. Baradaran Akbarzadeh**, W.C. Tai "*Analysis of an Electrically Interconnected Suspension System with various electrical network configurations Under Random or Bump Excitations*" (Paper accepted and pending for publication in *AMSE's IDETC-CIE 2025 conference*)
4) **M. Baradaran Akbarzadeh**, W.C. Tai "*Energy Regeneration of an Electrically Interconnected Suspension System with various electrical network configurations Under Random or Bump Excitations*" (Paper accepted and published in *Journal of Vibration and Acoustics*) Link
5) **M. Baradaran Akbarzadeh**, W.C. Tai, Ranjan Mukherjee "*Enhanced energy Conversion Using Piecewise-Constant Damping: A Fluid-Conveying Pipe Example*" (Paper accepted and pending for publication in *AMSE's IDETC-CIE 2026 conference*)
6) **M. Baradaran Akbarzadeh**, W.C. Tai, Ranjan Mukherjee "*Energy Harvesting Using Time-Modulated Damping: A Fluid-Conveying Pipe Example*" (Paper submitted in Journal of Sound and Vibration)

Conference Presentations

1) **M. Baradaran Akbarzadeh**, W.C. Tai "*Analysis of an Electrically Interconnected Half-Car Suspension System With Resistive Loads Under Random Road Excitation*" (Paper accepted and published in *AMSE's IDETC-CIE 2024 conference*)
2) **M. Baradaran Akbarzadeh**, W.C. Tai "*Analysis of an Electrically Interconnected Suspension System with various electrical network configurations Under Random or Bump Excitations*" (Paper accepted and pending for publication in *AMSE's IDETC-CIE 2025 conference*)
3) **M. Baradaran Akbarzadeh**, W.C. Tai, Ranjan Mukherjee "*Enhanced energy Conversion Using Piecewise-Constant Damping: A Fluid-Conveying Pipe Example*" (Paper accepted and pending for publication in *AMSE's IDETC-CIE 2026 conference*)

Honors and awards

- Achieved 1st rank in B.Sc. degree in the class of **2016**
- Student of the year in faculty of engineering in the academic year of **2014-2015**

Leadership & University Services

- President of the Mechanical Engineering Graduate Student Association (ME-GSA) at MSU **26-27**
- Advisory Committee of Mechanical Engineering Department at MSU **25-26**
- Engineering Research Committee College of Engineering at MSU **26-27**

References

Wei Che Tai, Associate Professor, Michigan State University, USA
Email: **taiweich@msu.edu**
Ranjan Mukherjee, Professor, Michigan State University, USA
Email: **mukherji@msu.edu**

Brian Feeny, Professor, Michigan State University, USA
Email: **feeny@egr.msu.edu**